

Analysis

Exon 1 methylation status of *CDH13* is associated with decreased overall survival and distant metastasis in patients with postoperative colorectal cancer

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Abstract

Background Cadherin 13 (CDH13) is a member of the cadherin superfamily that exerts tumor-suppressive effects on cancers derived from epithelial cells. Although hypermethylation of *CDH13* promoter has been reported in various cancers, its prognostic value for colorectal cancer (CRC) is still controversial. The methylation alterations of *CDH13* within exon 1 have not yet been investigated.

Methods A total of 49 CRC patients were recruited for the prospective study. The methylation status of CpG sites was quantified by Bisulfite Amplicon Sequencing (BSAS) in malignant tissues and adjacent normal tissues. The primary end-point of the study was overall survival (OS) after surgery. The relationship between methylation level with pathological stage and OS was also evaluated.

Results Compared with adjacent normal tissues, the overall average methylation level within exon 1 was significantly increased in tumor tissues ($p < 0.001$). The association study showed that the hypermethylation status of the CpG1 site was non-significantly associated with the presence of distant metastasis ($p = 0.032$). Moreover, the hypermethylation of two CpG sites, including CpG1 ($p = 0.003$) and CpG5 ($p = 0.032$), was associated with worse OS in CRC. Co-hypermethylation of CpG1 and CpG5 sites was significantly associated with a worse clinical outcome (HR: 4.43 [95% CI 1.27–15.46]; $p = 0.019$) in multivariate Cox regression analysis.

Conclusion The methylation level of *CDH13* exon 1 in CRC tissue was significantly higher than in adjacent normal tissues. Hypermethylation at the CpG1 site suggests a risk of distant metastasis in CRC. The hypermethylation of the CpG1 site and CpG5 site, including the co-hypermethylation of these two sites, may serve as a valuable prognostic biomarker.

Study Highlights

What is known • Cadherin 13 (CDH13) exerts tumor-suppressive effects on cancers derived from epithelial cells.

- Methylation of the exon 1, 5'untranslated regions (5' UTR) and 200 base pairs (bp) upstream the transcription start site (TSS200) are vital in gene methylation.
- The prognostic value of status of methylation of CDH12 for colorectal cancer (CRC) is still controversial.

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What is new here • The hypermethylation of the CpG1 site and CpG5 site, including the co-hypermethylation of these two CpG site, may serve as a valuable prognostic biomarker.

Keywords Colorectal cancer (CRC) · *CDH13* gene · Methylation · CpG site

1 Introduction

Colorectal cancer (CRC) is the third most common human malignant disease and was regarded as the second leading cause of cancer deaths worldwide in 2018 [1]. Although most patients undergo surgery with adjuvant chemotherapy and/or radiotherapy, the median overall survival (OS) for the most malignant CRC ranges from 4 to 6 months only [2]. Evidence indicates that alterations in epigenetic machinery, such as DNA methylation, are critical for cancer initiation and progression [3]. Hypermethylation in cytosine-phosphate-guanine (CpG) island regions of tumor suppressor gene usually affects important biological processes, interferes with DNA repair, regulates the cell cycle and promotes the formation of tumor cells, thus impacting tumor progression and prognosis [4, 5]. A series of methylated tumor suppressor genes such as transmembrane protein 240 (TMEM240), chondrolectin (CHODL) and ependymin related 1 (EPDR1) have been shown to have predictive value for CRC prognosis [6–8].

H-cadherin, encoded by the *CDH13* gene, is an atypical member of a large cadherin superfamily, devoid of a transmembrane domain and anchored to the exterior surface of the plasma membrane via a glycosylphosphatidylinositol anchor [9]. The dysfunction of *CDH13* has been linked with various diseases such as cancer, neurodevelopmental disorder and atherosclerosis [10]. Previous studies have indicated that loss of *CDH13* expression may account for aberrant cell adhesion, cell signal transduction and cell proliferation, impacting the survival, growth and transformation of tumor cells in numerous instances of tumorigenesis, such as gastric cancer and liver cancer [11, 12].

Currently, research on DNA methylation in CpG island is mainly focused on the promoter region and exon 1. Data showed that abnormal hypermethylation of DNA in CpG island can lead to transcriptional inactivation, resulting in decreased gene expression or even no expression, thus promoting the occurrence of malignant tumors [13]. The change in the methylation pattern of *CDH13* in tumor cells was first found in the region related to the promoter, so the previous studies on aberrant methylation of *CDH13* were mainly focused on the promoter [14]. The role of hypermethylation in the *CDH13* promoter region in CRC has been widely studied and most studies have found that CRC tumor tissues are more prone to hypermethylation than adjacent normal tissues [15–17]. Some studies reported that *CDH13* methylation is related to selected clinicopathological features or poor prognosis of CRC. Wang et al. (2012) further demonstrated that the promoter methylation of *CDH13* is related to poor differentiation and shorter survival time [18]. However, Duan et al. (2017) showed that the promoter methylation status of *CDH13* was closely related to the risk of CRC, but it was of limited value in evaluating the prognosis of CRC patients [19].

The relationship between *CDH13* promoter methylation and the prognosis of CRC patients remains unclear. It has been suggested that the prognostic value of methylation in other important regions of *CDH13* should be evaluated in CRC patients. Indeed, Ozer et al. (2017) proposed that exon 1, the 5' UTR, and the 200 bp region upstream of the transcription start site are more critical than other genomic regions in the internal mechanisms of cancer [20].

In the present study, we investigated methylation levels of *CDH13* exon 1 in a cohort of CRC patients by using the BSAS method. The DNA methylation of *CDH13* has been previously explored in colorectal cancer and other cancers. However, our study specifically focuses on exon 1 of *CDH13*, which has not been as extensively studied in relation to CRC prognosis. This provides a novel contribution by identifying hypermethylation at specific CpG sites within exon 1. While *CDH13* promoter methylation has been well-documented, our study offers new insights by targeting exon 1, which could serve as a distinct prognostic biomarker. We will elaborate on this in the introduction and discussion sections to better convey the unique contributions of our work. Furthermore, we aimed to assess CpG site-specific hypermethylation for accurately predicting prognosis biomarkers.

2 Materials and methods

2.1 Ethical statement

The present study was conducted in accordance with the Declaration of Helsinki and approved (approval no. 【2021】研预审第(03)号) by the Ethics Committee of the Shanghai East Hospital. Written informed consent was obtained from all related participants for sample collection and data analysis.

2.2 Study subjects and data collection

Samples from 49 patients (32 males and 17 females) with CRC confirmed by pathological diagnosis were collected from a follow-up study. These patients received surgical resection in Shanghai East Hospital from October 2009 to June 2010 and from July 2011 to March 2012, respectively. None of the patients had any other history of cancer or received pre-operative radiotherapy or chemotherapy prior to CRC surgery. Clinical data of age, gender, tumor markers, clinic pathologic characteristics, and clinical information about disease and treatment were collected from the medical record registration system.

2.3 Analysis flow

The experiment followed the following flow chart (Fig. 1), and performed strict quality testing. The methods and the DNA methylation standards used in our protocol were referring to the literature [21].

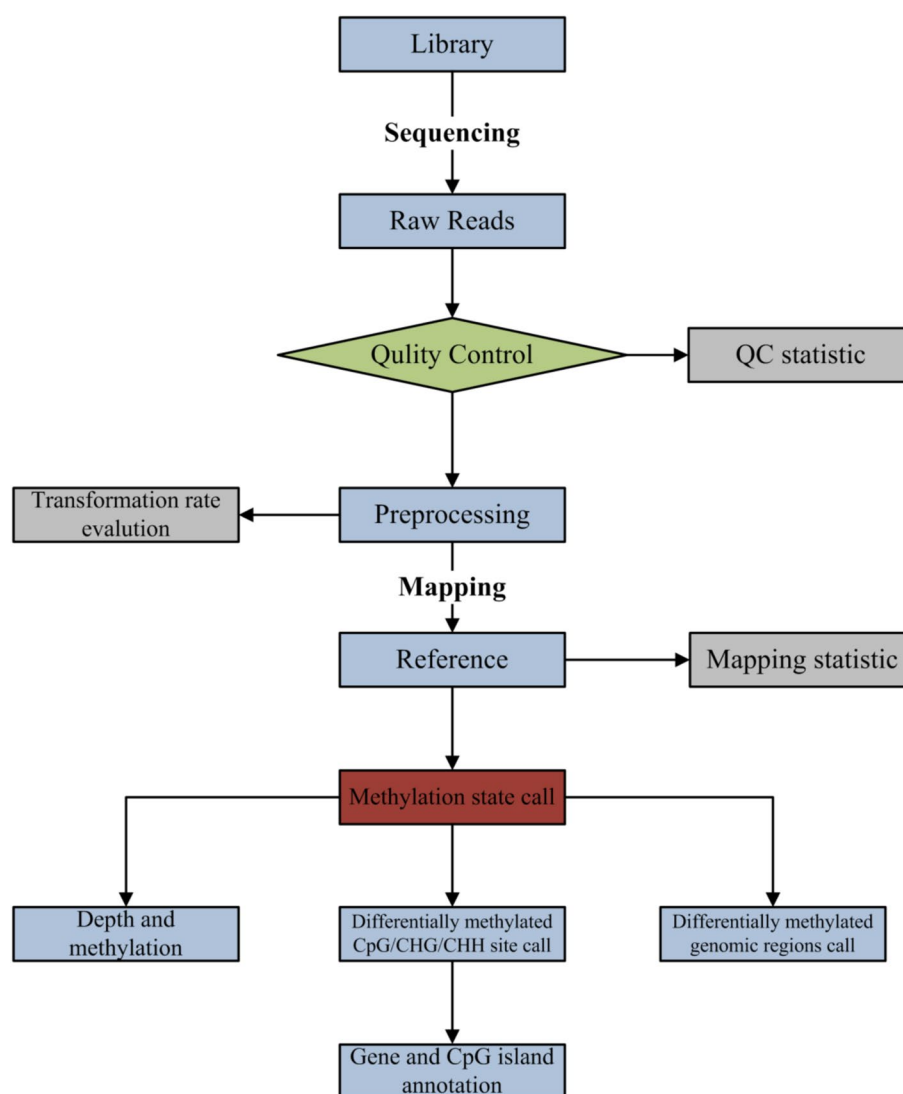
2.4 DNA isolation and bisulfite treatment

Genomic DNA was extracted from paraffin-embedded CRC tissues using standard phenol–chloroform extraction and ethanol precipitation protocol as previously described [22]. High-molecular-weight genomic DNA from CRC tumor samples and corresponding normal tissue samples were purified using the QIAmp DNeasy Blood & Tissue Kit (Qiagen, Hilden, Germany) according to the manufacturer's instructions. The yield and purity were determined using a Nanodrop 1000 Spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA). Bisulfite treatment was performed using EZ DNA Methylation-Gold TM Kit (ZYMO, California, USA). After bisulfite treatment, all unmethylated cytosine was converted to uracil while leaving methylated cytosine unchanged. Bisulfite-treated genomic DNA samples were stored at -20°C until use.

2.5 Primer design and DNA methylation analysis using BSAS

As shown in Fig. 2A, the *CDH13* exon 1 is located on chromosome 16 (GRCh38:Chr16: 82626969–82627137). The sequences of the target DNA fragment, which contains 13 CpG sites, were obtained from the genomic sequence (GRCh38:Chr16: 82626991–82627122). The purified bisulfite-modified DNA was amplified using the following primers containing 186 bp sequence which was designed by Meth Primer software online: *CDH13* gene forward, 5' TTTGGGAAGTTGGTTGGTTG-3 and reverse, 5'-ACAACCCCTCTCCCTACCT-3'. The modified DNA was amplified by polymerase chain reaction (PCR) using the following conditions: 95 °C for 5 min and then 45 cycles of 95 °C for 15 s, 55 °C for 20 s and 72 °C for 20 s, followed by a 5-min incubation at 72 °C. The PCR products were analyzed by gel electrophoresis in 1.5% agarose gel to confirm that a single band had been produced. Then, column purification and recovery were established according to the instructions of the protocol for FavorPrep GEL/PCR Purification Kit. And then a series of DNA methylation standards were constructed, which included 100%, 80%, 60%, 40%, 20% and 0% methylated DNA. Through the construction of a second-generation sequencing library, those amplified DNA was sequenced using the 2*300 of the Illumina Miseq platform. The original image data obtained by sequencing were converted into sequence data by base calling (raw data). After quality control, the qualified sequencing results

Fig. 1 Brief Analysis Flow Chart



were compared with the target fragment sequence by Bowtie 2 software. Target CpG sites were evaluated by using the Methyl Kit package of the R language, which converts the amplifier sequencing diagram to numerical values and calculates the proportion of DNA methylation at each base as a C/T ratio.

2.6 Statistical analysis

The output data includes site-specific percent DNA methylation for each CpG in the investigated area, as well as overall average percent DNA methylation, which was calculated as a mean value of all 13 CpG sites detected within exon 1. Overall or site-specific DNA methylation levels of *CDH13* were compared between CRC tissues and control subjects (the control group is the group with adjacent normal tissues from the same patients) using the non-parametric Wilcoxon's Rank Sum test. Receiver operator characteristic (ROC) curve analysis using MedCalc Software was constructed for distinguishing CRC and adjacent normal tissues. Univariate logistic regression analysis was performed to estimate the odds ratio (OR) of clinicopathological characteristics. The prognostic value of clinicopathological factors and *CDH13* site-specific hypermethylation was analyzed with Kaplan–Meier survival analysis and standard log-rank test. Multivariate models were then developed that adjusted for the most important covariates. All statistical analyses were performed using GraphPad Prism v8.3.0. $p < 0.05$ was considered statistically significant.

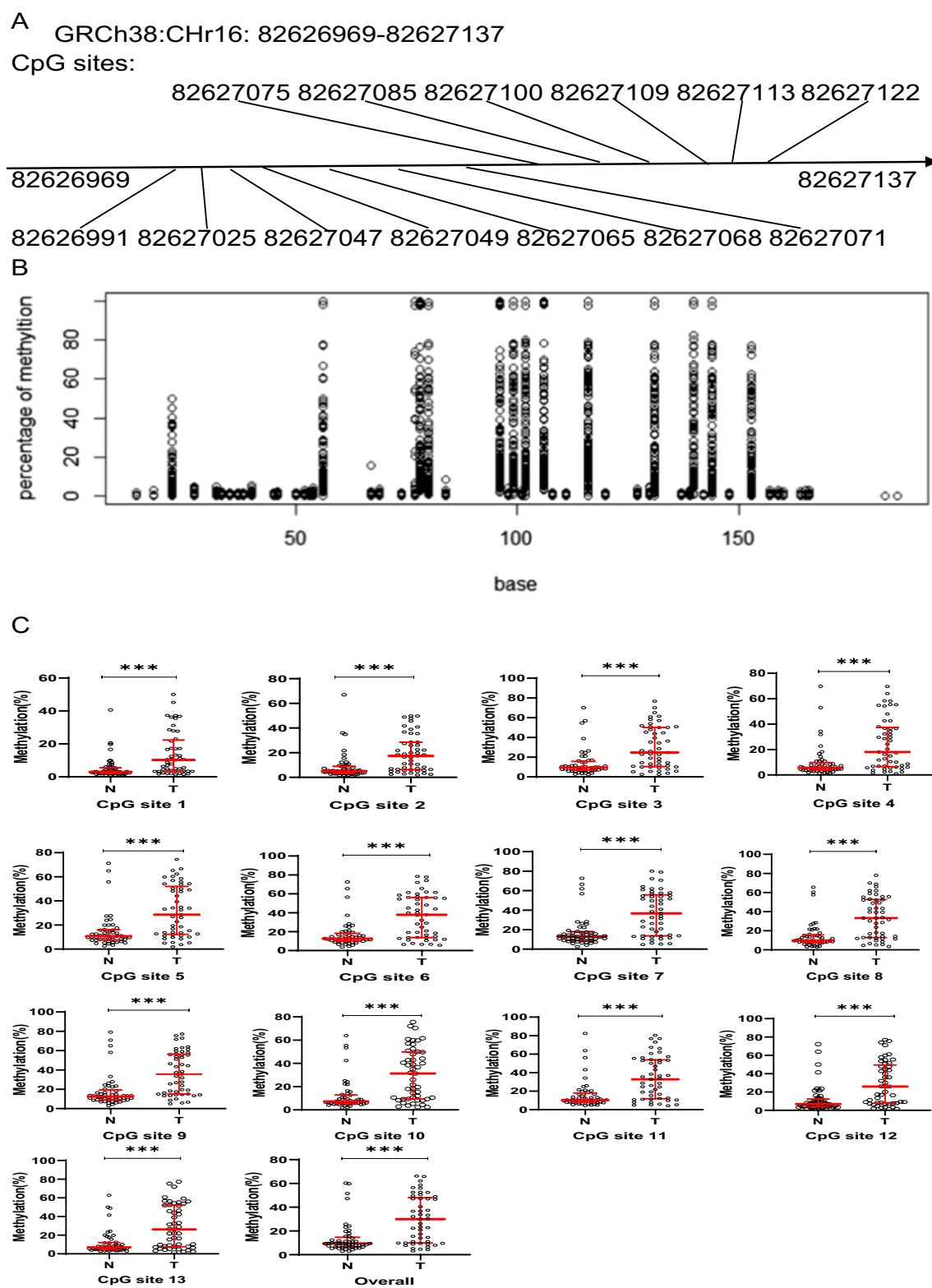


Fig. 2 Site-specific analysis of CDH13 gene exon1 by BSAS in colorectal tissues. **A** The position of 13 CpG sites in exon 1. **B** Methylation levels at 186 bp region. **C** Methylation levels of each site and overall average level are represented. Horizontal lines represent the median and the 25th–75th percentile. T colorectal cancer tissues, N adjacent normal tissues. *** $p < 0.001$

3 Results

3.1 Clinicopathological characteristics of all 49 sample

According to the data format, the clinicopathological characteristics of all the samples are listed in Table 1.

3.2 Overall average and site-specific DNA methylation levels of *CDH13* exon 1 in CRCs.

Using the BSAS method, we quantified the DNA methylation status of 13 individual CpG sites located in the *CDH13* exon 1. The aim of this analysis was to define the methylation levels of the CRC patient's tumor and adjacent normal tissue and to assess whether any of the tested sites could be considered as a potential tumor-tissue marker. As shown in Fig. 2B, methylation level in all samples is presented, and percent methylation at 13 CpG sites, as well as overall average percent methylation, are significantly higher in CRCs than in comparable normal tissues, as shown in the following Table 2 and Fig. 2C.

ROC curve analysis was performed to evaluate the diagnostic value of differentially methylated CpG sites. As shown in Fig. 3, 13 CpG sites showed a moderate AUC of the ROC curves, ranging from 0.729 to 0.776. Since an AUC of ≥ 0.75 generally indicates a marker that could potentially have clinical utility [23], we combined these 13 CpG sites to improve the diagnostic power. The result indicated that a combination of these indicators increased sensitivity and specificity (AUC = 0.844; $p < 0.0001$). Age-dependent DNA methylation has been recognized as an important physiological process [24, 25]; In this study, we included age in the diagnostic model, and the result revealed no significant difference in diagnostic performance compared with the combined diagnosis of the 13 CpG sites. This suggests that the DNA methylation status of specific CpG sites in *CDH13* are not promising diagnostic biomarker candidates for the differentiation of CRC tissues from surrounding non-malignant tissue.

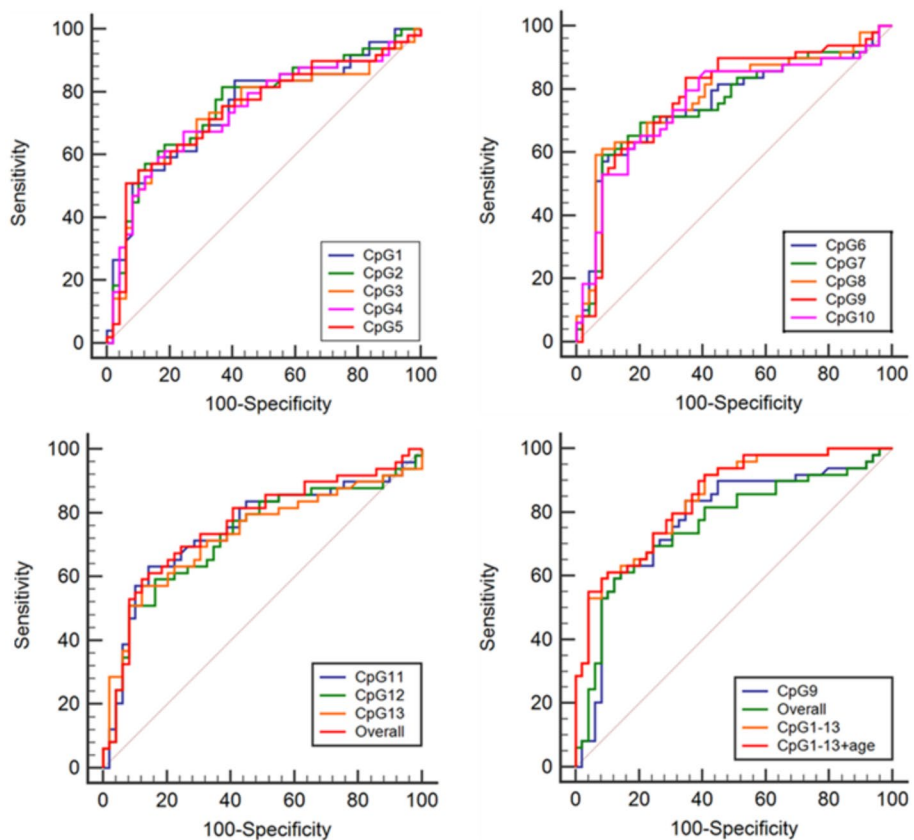
Table 1 Baseline characteristics of 49 patients in the study

Variable	Classification	Number	Frequency (%)
Age (yrs)	< 65	24	51.0
	≥ 65	25	49.0
Gender	Male	32	65.3
	Female	17	34.7
Tumor location	Colon	24	51.0
	Rectum	25	49.0
Differentiation	High-Moderate	37	75.5
	Moderate-Low	12	24.5
Tumor stage	T1	3	6.1
	T2	4	8.2
	T3	30	61.2
	T4	12	24.5
Lymph node metastasis	N0	25	51.0
	N1	14	28.6
	N2	10	20.4
Distant metastasis	M0	43	87.8
	M1	6	12.2
Tumor stage (TNM)	I	6	12.2
	II	17	34.8
	III	20	40.8
	IV	6	12.2
Vascular vessel permeation	Absent	28	57.1
	Present	21	42.9
Nerve invasion	Absent	32	65.3
	Present	17	34.7

Table 2 Methylation level in all samples

	Tumor	Normal	P
Site1	10.25(3.39–22.30)	2.99(2.06–5.52)	<0.001
Site2	17.36(6.24–28.53)	5.07(3.27–8.98)	<0.001
Site3	24.68(10.43–50.00)	9.38(6.65–15.62)	<0.001
Site4	18.05(6.48–37.37)	5.60(3.65–9.72)	<0.001
Site5	28.57(12.24–51.99)	10.75(7.62–16.02)	<0.001
Site6	37.87(13.71–56.12)	12.59(9.97–18.20)	<0.001
Site7	36.69(13.61–55.67)	13.02(8.78–17.87)	<0.001
Site8	33.26(12.91–52.70)	9.73(7.48–15.65)	<0.001
Site9	35.64(15.07–56.28)	12.45(8.99–19.21)	<0.001
Site10	31.28(9.34–49.81)	7.29(5.30–12.90)	<0.001
Site11	32.76(11.73–53.74)	10.37(7.80–17.84)	<0.001
Site12	25.98(7.78–49.44)	6.84(3.91–12.40)	<0.001
Site13	26.12(7.63–52.07)	6.99(4.60–12.07)	<0.001
Overall	29.93(9.85–48.04)	9.50(6.41–14.62)	<0.001

Fig. 3 ROC curve analysis was performed to evaluate the diagnostic value of differentially methylated CpG sites



3.3 Association between DNA methylation level and clinicopathological characteristics in CRCs.

Although the average DNA methylation level in adjacent normal tissue was low in exon 1, it could be far beyond the average in some cases. Even in a few samples, the methylation levels in non-malignant tissues were higher than the cutoff values derived from the ROC curve analysis. This non-negligible methylation pattern suggests that a criterion must first be set up to define hypermethylation at CpG sites in order to analyze the association of *CDH13* hypermethylation with clinicopathological characteristics. We set a threshold value of 25% DNA methylation difference between

tumor and non-malignant to ensure and only aberrant hypermethylation cases were assigned as positive. Next, the relationship between the methylation status of exon 1 and clinicopathological features was assessed by univariate logistic analysis. The findings are shown in Table 3. A positive correlation was observed between hypermethylation at CpG site 1 and the presence of distant metastasis, which was statistically significant (OR: 7.60 [95% CI 1.19–48.44], $p = 0.032$). Considering that both site 1 and site 5 indicate poor prognosis, which is shown in Fig. 4M and N, we

Table 3 Univariate associations of site-specific and overall methylation status of exon 1 in colorectal cancer with clinicopathological characteristics (odds ratio and 95% CI)

	Site 1 ^a	Site 2 ^a	Site 3 ^a	Site 4 ^a	Site 5 ^a
Gender ^b	1.73 (0.31–9.68)	1.56 (0.35–6.84)	1.62 (0.48–5.44)	1.87 (0.53–6.55)	1.43 (0.42–4.81)
Age ^c	0.28 (0.06–1.35)	1.02 (0.25–4.12)	1.29 (0.40–4.22)	0.99 (0.30–3.27)	1.65 (0.49–5.52)
Tumor location ^d	0.29 (0.05–1.60)	0.83 (0.22–3.20)	0.46 (0.15–1.47)	0.45 (0.14–1.45)	0.38 (0.12–1.24)
Tumor size ^e	0.70 (0.15–3.30)	1.65 (0.43–6.37)	2.40 (0.75–7.65)	2.38 (0.73–7.69)	1.42 (0.45–4.46)
Differentiation ^f	2.13 (0.43–10.69)	1.21 (0.26–5.54)	1.47 (0.40–5.43)	1.85 (0.49–6.89)	1.64 (0.44–6.10)
CEA ^g	0.86 (0.19–3.93)	0.68 (0.18–2.60)	0.95 (0.31–2.96)	0.97 (0.31–3.07)	1.14 (0.36–3.58)
CA19-9 ^h	1.86 (0.38–9.20)	1.05 (0.23–4.76)	1.20 (0.34–4.30)	1.52 (0.42–5.48)	1.35 (0.38–4.84)
Tumor invasion ⁱ	1.44 (0.49–4.25)	0.59 (0.26–1.36)	1.33 (0.43–4.16)	0.93 (0.44–1.93)	0.92 (0.44–1.90)
Lymph node metastasis ^j	3.83 (0.69–21.30)	2.16 (0.54–8.64)	1.27 (0.41–3.94)	1.80 (0.56–5.75)	1.50 (0.48–4.73)
Distant metastasis ^k	7.60 (1.19–48.44)*	1.89 (0.30–12.01)	3.06 (0.50–18.58)	3.73 (0.61–22.80)	3.38 (0.55–20.55)
Tumor stage ^l	8.11 (0.91–71.94)	2.96 (0.68–12.91)	1.04 (0.50–2.16)	1.96 (0.60–6.35)	1.61 (0.51–5.09)
Vascular vessel permeation	1.41 (0.31–6.45)	1.15 (0.30–4.42)	1.00 (0.32–3.14)	1.35 (0.42–4.30)	1.16 (0.37–3.66)
Nerve invasion ⁿ	2.15 (0.46–9.99)	1.10 (0.27–4.45)	0.90 (0.27–2.96)	1.17 (0.35–3.88)	1.02 (0.31–3.38)
	Site 6	Site 7	Site 8	Site 9	Site 10
Gender ^b	2.09 (0.63–6.91)	2.36 (0.70–7.95)	1.84 (0.56–6.05)	1.83 (0.55–6.16)	2.08 (0.62–6.99)
Age ^c	0.85 (0.27–2.74)	0.94 (0.29–2.97)	1.07 (0.33–3.41)	1.47 (0.45–4.80)	0.82 (0.26–2.64)
Tumor location ^d	0.79 (0.26–2.42)	0.56 (0.18–1.74)	0.67 (0.22–2.05)	0.39 (0.12–1.25)	0.66 (0.21–2.04)
Tumor size ^e	1.56 (0.50–4.85)	2.10 (0.67–6.60)	1.29 (0.42–4.00)	1.46 (0.47–4.52)	1.25 (0.40–3.87)
Differentiation ^f	1.33 (0.36–4.95)	1.65 (0.44–6.15)	1.48 (0.40–5.51)	2.05 (0.55–7.70)	1.18 (0.32–4.33)
CEA ^g	1.49 (0.48–4.60)	1.09 (0.36–3.35)	1.27 (0.41–3.92)	1.11 (0.36–3.45)	0.94 (0.30–2.88)
CA19-9 ^h	1.60 (0.44–5.84)	1.30 (0.37–4.65)	0.77 (0.22–2.74)	1.63 (0.46–5.85)	1.46 (0.41–5.21)
Tumor invasion ⁱ	1.13 (0.55–2.32)	1.01 (0.49–2.07)	0.87 (0.42–1.79)	1.03 (0.50–2.12)	1.17 (0.56–2.42)
Lymph node metastasis ^j	1.09 (0.36–3.35)	1.08 (0.35–3.32)	0.67 (0.22–2.05)	1.50 (0.48–4.65)	1.77 (0.57–5.51)
Distant metastasis ^k	1.91 (0.32–11.55)	2.30 (0.38–13.92)	0.96 (0.17–5.27)	2.78 (0.46–16.84)	2.53 (0.42–15.30)
Tumor stage ^l	1.07 (0.35–3.29)	1.09 (0.36–3.35)	0.66 (0.21–2.04)	1.56 (0.50–4.85)	1.82 (0.58–5.67)
Vascular vessel permeation	1.86 (0.59–5.93)	1.78 (0.57–5.58)	1.10 (0.36–3.41)	1.70 (0.54–5.34)	1.05 (0.34–3.26)
Nerve invasion ⁿ	1.43 (0.44–4.69)	1.28 (0.39–4.14)	1.62 (0.49–5.32)	1.64 (0.50–5.38)	1.45 (0.44–4.71)
	Site 11	Site 12	Site 13	Overall	Sites 1&5
Gender ^b	1.43(0.44–4.69)	1.83(0.55–6.16)	2.12(0.61–7.42)	2.40(0.69–8.40)	1.39(0.24–8.05)
Age ^c	1.17(0.34–3.76)	1.03(0.32–3.31)	1.65(0.49–5.52)	1.29(0.40–4.22)	0.38(0.07–1.91)
Tumor location ^d	0.66(0.21–2.04)	0.55(0.18–1.73)	0.38(0.12–1.24)	0.32(0.10–1.06)	0.14(0.02–1.25)
Tumor size ^e	1.25(0.40–3.87)	1.46(0.47–4.52)	1.01(0.32–3.17)	1.70(0.54–5.34)	0.91(0.18–4.57)
Differentiation ^f	1.18(0.32–4.33)	1.31(0.36–4.84)	0.66(0.17–2.57)	1.47(0.40–5.43)	2.75(0.52–14.59)
CEA ^g	0.94(0.30–2.88)	0.80(0.26–2.47)	0.81(0.26–2.55)	0.95(0.31–2.96)	0.62(0.12–3.12)
CA199 ^h	0.96(0.27–3.42)	1.07(0.30–3.83)	1.35(0.38–4.84)	1.83(0.51–6.59)	1.13(0.19–6.67)
Tumor invasion ⁱ	1.02(0.49–2.09)	1.03(0.50–2.12)	0.92(0.44–1.90)	0.91(0.44–1.87)	1.13(0.39–3.27)
Lymph node metastasis ^j	1.27(0.41–3.92)	1.50(0.48–4.65)	1.07(0.34–3.35)	1.27(0.41–3.94)	3.03(0.53–17.39)
Distant metastasis ^k	2.53(0.41–15.30)	2.78(0.46–16.84)	3.38(0.55–20.55)	3.06(0.50–18.58)	9.75(1.46–65.36)*
Tumor stage ^l	1.30(0.42–4.01)	1.56(0.50–4.85)	1.14(0.36–3.58)	1.33(0.43–4.16)	6.60(0.73–59.68)
Vascular vessel permeation	1.05(0.34–3.26)	0.87(0.28–2.71)	1.16(0.37–3.66)	1.41(0.45–4.41)	1.00(0.20–5.04)
Nerve invasion ⁿ	1.01(0.31–3.27)	1.14(0.35–3.72)	1.02(0.31–3.38)	1.30(0.40–4.25)	1.50(0.29–7.65)

Table 3 (continued)^a $p < 0.05$ ^aOR, odds ratio with 95% confidence interval^bgender (male; female)^cage (< 65 years; $\geq$ 65 years)^dtumor location (colon; rectum)^eTumor size ($\leq$ 5 cm; > 5 cm)^fDifferentiation (well-moderate, moderate-poor)^gCEA ($\leq$ 5 ng/ml; > 5 ng/ml)^hCA199 ($\leq$ 27 ng/ml; > 27 ng/ml)ⁱtumor invasion (T1 – T2; T3 – T4)^jlymph node metastasis (yes vs no)^kdistant metastasis (yes vs no)^ltumor stage (I – II; III – IV)^mvascular vessel permeation (yes vs no)ⁿnerve invasion (yes vs no)

CA cancer antigen, CEA carcinoembryonic antigen

combined them to integrally analyze their relationship between the DNA methylation status and clinicopathological features. A significant correlation between the co-hypermethylation of sites 1 and 5, and distant metastasis was observed (OR: 9.75 [95% CI 1.46 – 65.36]; $p = 0.019$) in Table 3.

3.4 Effect of *CDH13* hypermethylation on poor survival of CRC patients

The last follow-up date for this study was April 19, 2020, with an average follow-up time of 116 months. During the follow-up period, survival data of 49 patients were obtained, of whom 27 (55.1%) died. The time from patient's surgery to death for various reasons or the last follow-up visit was defined as the overall survival (OS) time. With the cutoff values described above, we used Kaplan–Meier survival curves and a log-rank test to determine the effect of *CDH13* hypermethylation on the survival of CRC patients. In accordance with the findings shown in Fig. 4, the hypermethylation versus non-hypermethylation status of site 1 (OS, 52.0 ± 12.6 months vs 94.7 ± 5.6 months on average, $p = 0.003$) and site 5 (OS, 79.9 ± 7.3 months vs 97.0 ± 8.2 months on average, $p = 0.032$) was associated with poor prognosis (Fig. 4M and N), but no significant correlation was found in the other sites (Fig. 4A–L). Considering the statistical significance between tumorous hypermethylation and OS was only found in these two sites, we combined them to integrally analyze their contribution to patients' outcomes. When we combined patients with hypermethylation at both sites, they achieved significantly worse survival than the patients without hypermethylation at both sites (57.0 ± 13.4 months vs 92.8 ± 5.8 months on average, $p = 0.020$) (Fig. 4O).

The prognostic value of clinicopathological factors and hypermethylation were analyzed by univariate and multivariate Cox regression analysis. The multivariate Cox analysis included CEA, TNM stage, lymph node metastasis and distant metastasis. To answer the question of whether the co-hypermethylation status of site 1 and site 5 contributes independently prognosis evaluation or is associated with disease factors such as CEA, TNM stage, lymph node metastasis and distant metastasis, the relationship of site-specific hypermethylation with OS was studied using multivariate Cox regression analysis. Results indicated that co-hypermethylation of sites 1 and 5 is a predictor of poor prognosis for CRC patients both in univariate analysis (HR:2.82 [95%CI 1.13–7.04]; $p = 0.027$) and multivariate Cox regression analysis (HR: 4.43 [95% CI 1.27–15.46]; $p = 0.019$) (Table 4).

4 Discussion

DNA methylation plays a key role in CRC and has been confirmed as a new prognostic factor of the disease [26]. Previous studies used methods that just qualitatively analyze particular CpG sites or quantitatively analyze overall DNA methylation levels of the promotor of the candidate gene. Due to the inconsistent research results, its correlation between DNA methylation status and clinicopathological significance remains controversial. Currently, DNA

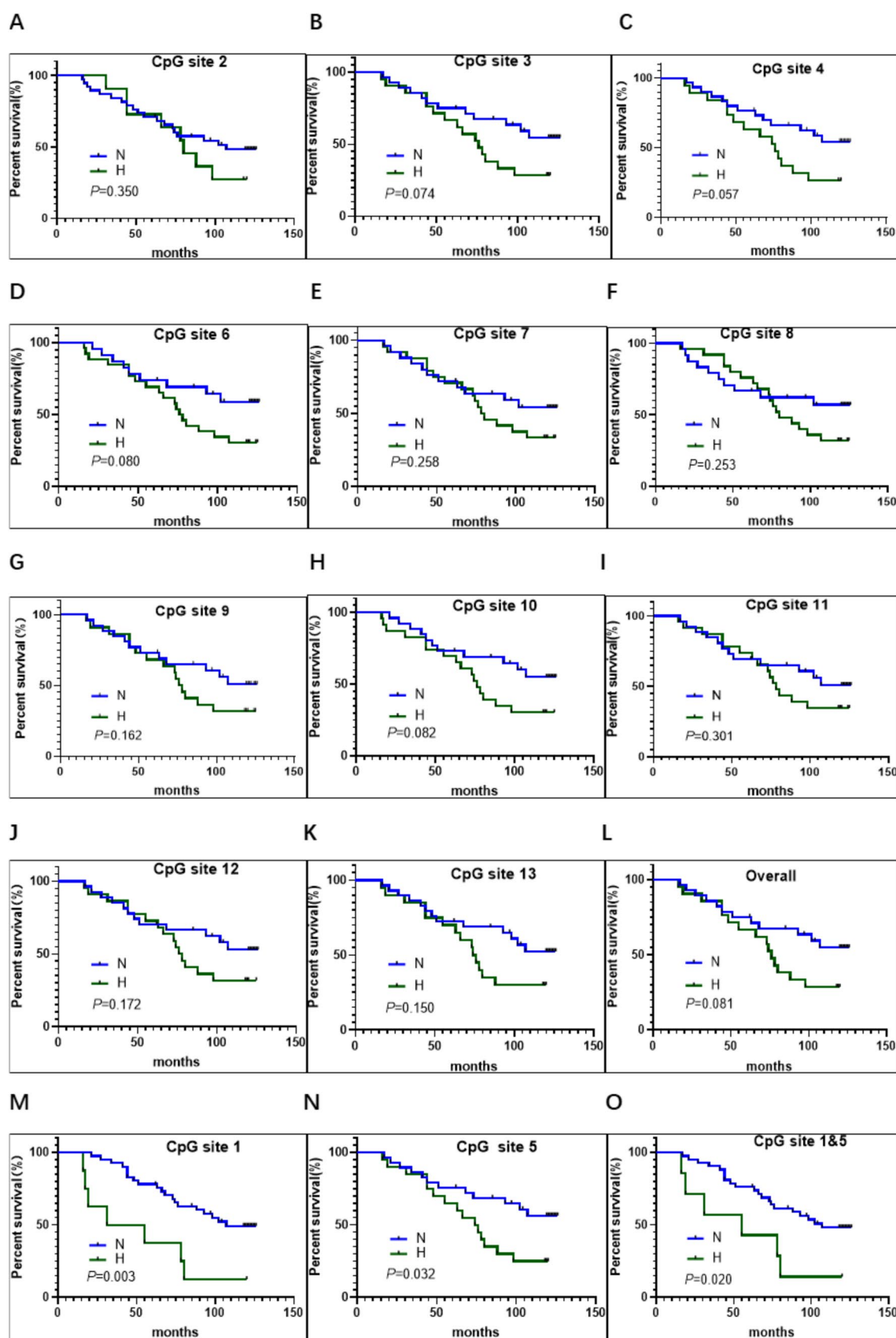


Fig. 4 Kaplan-Meier survival curves for overall survival according to the threshold value for prognostic DNA methylation sites. *N* non-hypermethylation, *H* hypermethylation

Table 4 Prognostic value of clinicopathological factors and co-hypermethylation of site 1 and site 5 in univariate and multivariate Cox regression analysis (n = 49)

Variable	Univariate analysis		Multivariate analysis	
	HR		HR	
	(95% CI)	p	(95% CI)	p
CEA				
≤ 5 ng/ml	1.00 (reference)		1.00 (reference)	
> 5 ng/ml	3.95 (1.66 – 9.40)	0.002	5.07 (1.72 – 14.98)	0.003
Lymph node metastasis				
Absent	1.00 (reference)		1.00 (reference)	
Present	2.92 (1.30 – 6.55)	0.009	1.35 (0.19 – 9.67)	0.766
Distant metastasis				
Absent	1.00 (reference)		1.00 (reference)	
Present	16.10 (4.80 – 53.94)	< 0.001	4.55 (1.09 – 18.93)	0.037
Tumor stage				
I – II	1.00 (reference)		1.00 (reference)	
III – IV	4.08 (1.71 – 9.73)	0.002	2.36 (0.26 – 21.19)	0.444
Methylation status of sites 1 & 5				
Non-hypermethylation	1.00 (reference)		1.00 (reference)	
Co-hypermethylation	2.82 (1.13 – 7.04)	0.027	4.43 (1.27 – 15.46)	0.019

CEA carcinoembryonic antigen

methylation alteration in exon has attracted more attention since the CpG site was found to frequently occur in some tumor suppressor genes and DNA methylation in transcribed regions was also correlated with gene expression [27]. DNA methylation of exon-encoding regions on alternative splicing has been confirmed [28]. Studies have shown that most of the differential DNA methylation regions are located in the promoter regions near the transcriptional initiation site, 5' UTR and exon 1[29]. Li et al. (2018) have shown that it is the DNA methylation of specific CpG sites, and not the DNA methylation of the whole gene, is associated with clinical features and prognostic value, suggesting that site-specific DNA methylation has different biological functions [30]. The analysis of exon regions would provide more detectable sites and may be novel biomarkers or therapeutic targets in cancer [27].

To our knowledge, this is the first study where the methylation status of *CDH13* in CRC was evaluated using next-generation sequencing (NGS). BSAS is one of the most accurate methods available for quantifying DNA methylation at specific CpG sites within the target region of interest [31]. It is a sensitive and highly reproducible method that is uniquely suited to the analysis of a small fraction of the total DNA in clinical samples [32]. Our data showed the overall DNA methylation level of *CDH13* exon 1 and DNA methylation levels at all 13 CpG sites were significantly higher in CRC than in paired adjacent normal tissues, which was in line with a previous systematic study about tumor suppressor gene sequencing [7, 27].

We further explored the relationship between the aberrant hypermethylation of the *CDH13* site and clinicopathological features of CRC patients. However, our research revealed that *CDH13* methylation might occur in non-neoplastic tissues; a threshold should be set to confirm positive hypermethylation. Some groups have used median as a basis for cutoff value determination for prognostic DNA methylation markers in colon adenocarcinoma [8, 28, 33]. However, this may be suboptimal because the existence of outliers may have a considerable impact on the distribution of DNA methylation data in non-malignant tissues. Wang et al. (2016) set a threshold value derived from ROC curve analysis for hypermethylation positivity [34], often statistically optimal, but in our study, it is less compatible with the biological relevance. Ozer et al. (2017) found that 25% methylation change in previously determined genomic regions showed the highest inverse correlation with expression data [20]. Therefore, we set a threshold value at 25% to ensure that only aberrant hypermethylation cases were assigned as positive. The subsequent statistical analysis showed a significant correlation between *CDH13* exon 1 and the presence of distant metastasis, which suggested that the hypermethylation of *CDH13* was associated with CRC invasion and metastasis. Dmitrijeva et al. (2018) found that there may be CpG hypermethylation overlap sites in the process of aging and tumor [24]. No significant correlation was observed between age and hypermethylation at the CpG site, which may be attributed to various factors such

as the small sample size, the tumor's location, or the characteristics of the selected population. This suggests that the underlying mechanisms linking hypermethylation at CpG sites and aging require further investigation.

To analyze the association between CDH13 hypermethylation and cancer-related mortality, we conducted Kaplan–Meier survival analyses and univariate Cox proportional hazard regression to assess the impact of CDH13 hypermethylation on the survival of CRC patients. Our results indicated no significant association or predictive value for overall hypermethylation in relation to patient outcomes. However, the methylation status of the CpG1 and CpG5 sites was associated with a poor prognosis. To further determine whether co-hypermethylation at CpG1 and CpG5 serves as an independent prognostic predictor in patients following surgical resection of CRC, we performed multivariate Cox regression analysis. The results revealed that distant metastasis and CEA levels were independent predictors of CRC prognosis, consistent with findings from previous studies [35, 36].

Notably, co-hypermethylation of site 1 and site 5 appears to be an independent predictor of poor prognosis in CRC patients. This suggests that site-specific DNA methylation analysis provides more detailed information and offers researchers the opportunity to gain additional insights. The hypermethylation of CDH13 exon 1 may therefore serve as a valuable biomarker for assessing clinical outcomes in CRC patients.

5 Conclusion

Our research identified differentially methylated CpG sites in exon 1 of CDH13. The DNA methylation status of CpG1 and CpG5 could serve as promising and reliable prognostic biomarkers. However, given the limited published data on this topic, our findings should be validated in larger cohorts to confirm their clinical significance.

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Data availability The original data of DNA methylation at CpG that supports the findings of this study is available from the corresponding author on reasonable request and other data is provided within the manuscript (Figures and Tables).

Declarations

Ethics approval and consent to participate The present study was conducted in accordance with the Declaration of Helsinki and approved (approval no. 【2021】研预审第(03)号) by the Ethics Committee of the Shanghai East Hospital.

Competing interests The authors declare no competing interests.

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